

NAME

kh – compute the vapor-liquid constant

LIBRARY

iapws library (*libiapws*, *-liapws*)

SYNOPSIS

```

Fortran      pure subroutine kd(T,gas,heavywater,k)
               real(dp), intent(in), contiguous :: T(:)
               character(len=*), intent(in) :: gas
               integer(int32), intent(in) :: heavywater
               real(dp), intent(out), contiguous :: k(:)

C            void iapws_g704_kd(double *T,char *gas,int heavywater,double *k,int size_gas,size_t
               size_T)

Python      kd(T:np.ndarray,gas:str,heavywater:bool=False)->Union[np.ndarray,float]:

```

DESCRIPTION

Computes the Henry constant kH in MPa for a given temperature T . Compute the vapor-liquid constant kD for a given temperature T .

Arguments:

<i>T</i> (:)	Temperature in K.
<i>gas</i>	Gas.
<i>heavywater</i>	Flag if D2O (1) is used or H2O (0).
<i>k</i> (:)	Vapor-liquid constant in MPa.

RETURN VALUE

k is filled with NaN if gas is not found.

EXAMPLES

Fortran:

```

use iapws
real(dp) :: T(1), kD(1)
T(1) = 25.0_dp + 273.15_dp
call kd(T, "O2", 0, kD)

```

C:

```

#include "iapws.h"
double T = 25.0 + 273.15;
double kD;
iapws_g704_kd(&T, "O2", 0, &kD, strlen(gas), 1);

```

Python:

```

import numpy as np
import pyiapws

```

```
T = np.asarray([25.0])
kD = pyiapws.kd(T+273.15, "02", False)
print(kD)
```

SEE ALSO

iapws(3) **iapws_cli(1)** **iapws_kh(3)** **ciaaw(3)**, **codata(3)**,